MAHANDAR KUMAR

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PROFESSIONAL SUMMARY

BS Computer Science undergraduate (CGPA: 3.95) with strong foundations in Competitive Programming, AI application development, and full-stack software engineering. Experienced in building Retrieval-Augmented Generation (RAG) systems, AI chatbots, and scalable web applications using Python, C++, React, Next.js, and Node.js. Passionate about solving complex engineering problems through clean, efficient, and production-oriented software solutions.

CORE COMPETENCIES

Competitive Programming: Codeforces (Div 2/3/4 rated), Leetcode, cp-algorithms, greedy, simulation, cycle detection, constructive algorithms.

Languages: C++ (advanced), Python, JavaScript/TypeScript, SQL.

AI / ML & Backend: Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Prompt Engineering, ChromaDB, FAISS, Groq API, Ollama, Node.js.

Frontend & Tools: Next.js, React, React Three Fiber, Streamlit, Git/GitHub, Vercel.

PROJECTS & TECHNICAL EXPERIENCE

Multimodal RAG System from Scratch

Python · Retrieval-Augmented Generation · Vector Embeddings

- Built a retrieval-augmented generation pipeline from first principles that ingests research-paper text, generates embeddings, and retrieves contextually relevant passages to ground LLM-generated answers.

WhatsApp RAG Chatbot with Personal-Fact Memory

Node.js (Baileys) · Python (LangChain) · Groq API · ChromaDB/FAISS

- Built a WhatsApp chatbot pairing a Node.js messaging layer with a Python RAG backend, retrieving answers from a custom knowledge base over a persistent on-disk vector store. Implemented automatic extraction and natural-language recall of user-specific facts (name, location, occupation, interests) to personalize ongoing conversations.

RAG AI Chatbot (Text-Based)

LangChain · HuggingFace · ChromaDB · ChatGroq (Llama 3.1) · Streamlit

- Built a retrieval-augmented chatbot using LangChain, HuggingFace sentence-transformer embeddings, an in-memory ChromaDB vector store, and Llama-3.1-8B via the Groq API, served through a custom Streamlit interface.

Udaan - 3D Travel Planner Web App *Full-Stack Project*

Next.js 14 · React Three Fiber · Express/TypeScript · SQLite

- Built a full-stack travel-planning application with an Express/TypeScript API and a Next.js 14 frontend, featuring an interactive scroll-driven 3D animation built with React Three Fiber.

TypeMaster — Console Typing Speed Game *C++ (OOP)*

- Built a console-based typing speed game in C++, applying inheritance, polymorphism, and encapsulation to structure gameplay logic and scoring.

EDUCATION

Bachelor of Science in Computer Science (BSCS) *Expected 2029*

FAST National University of Computer & Emerging Sciences (FAST-NUCES) - Karachi, Pakistan | CGPA: 3.95

- Coursework and independent study in Data Structures & Algorithms, Object-Oriented Programming, and AI/ML fundamentals.

COMPETITIVE PROGRAMMING PROFILE

- Codeforces: _gazzy284 - structured progression through Div 3/4 → Div 2 problem sets.
- LeetCode: struck_high - ongoing problem-solving practice.

ADDITIONAL EXPERIENCE

AI Intern - FlyRank *Ongoing, Remote*

- Ongoing remote internship building applied AI fluency - practical use of AI tools and workflows on real projects.

ILM KI ROSHNI - Community Engagement Project *Volunteer*

- Contributed to a volunteer initiative teaching underprivileged students; authored the accompanying report and proposal documentation.